

$$B_{\text{resultant}} = B_S - B_C = 3.4 \times 10^{-6} \text{ T}$$

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E.m.f. is induced in the part of the length of the rod that forms a closed loop with the square. This length first increases than decreases, thus the e.m.f. induced will first increase, then decrease.

OR: Part of the rod forms a closed loop with the square. Magnetic flux linkage with this loop changes as the rod moves. The e.m.f. induced in the loop is proportional to the rate of increase of the area of the loop. This rate of increase of loop area first increases then decreases. Thus induced e.m.f. increases, then decreases.

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Power is independent of frequency. The area under the power-time remains the same regardless of frequency.